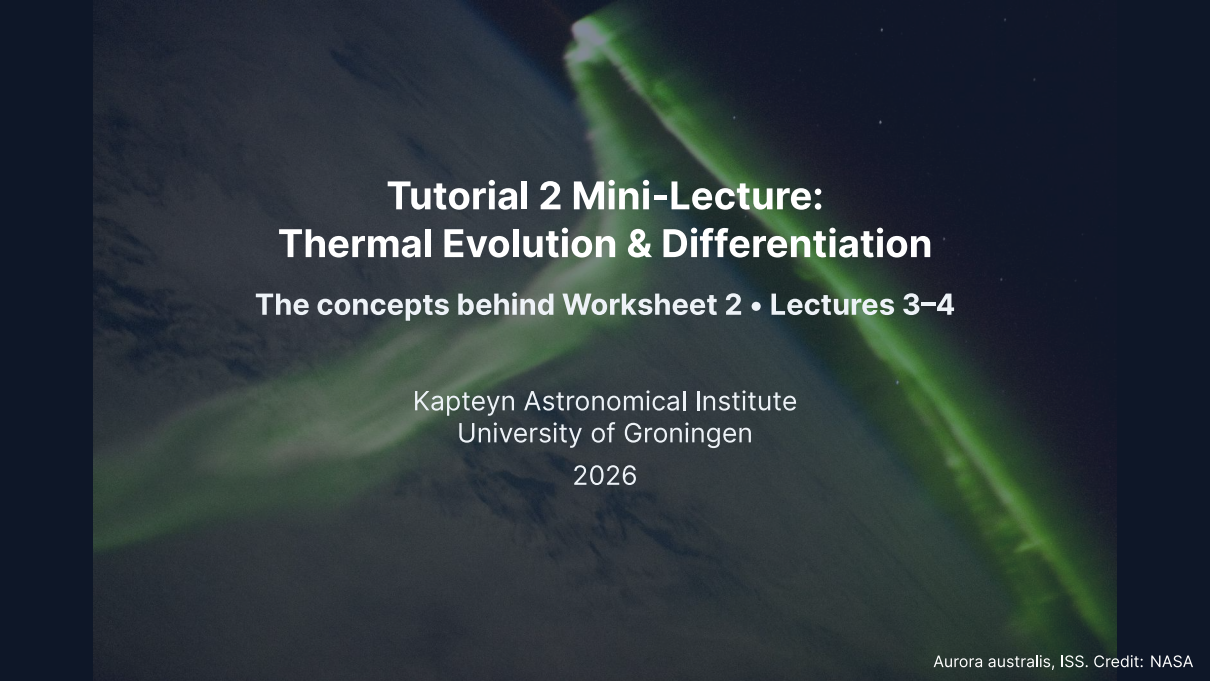


A photograph of the aurora australis (southern lights) as seen from the International Space Station (ISS). The image shows vibrant green and blue light curtains flowing across the dark, textured surface of the Earth's atmosphere. The aurora is composed of multiple parallel bands and rays, creating a dynamic and ethereal scene.

Tutorial 2 Mini-Lecture: Thermal Evolution & Differentiation

The concepts behind Worksheet 2 • Lectures 3–4

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2026

What Worksheet 2 trains

Five problems, all built on Lectures 3–4. The physics you need, per problem:

1. **The radiogenic clock:** isotope table to heating power, and the two heating eras.
2. **Conduction fails, convection takes over:** $\tau = L^2/\kappa$, Ra, the $\Delta T^{4/3}$ flux law.
3. **A minimal thermal evolution model:** heat in minus heat out, on paper and from its solution curve.
4. **Building a core and dating it:** partitioning, the late veneer, Hf-W ages.
5. **Dynamos and shields:** standoff distance, ohmic decay.

Every problem has the shape and level of one exam question. No part needs a computer.

The radiogenic clock

Specific heating = concentration $\times$ heat per kg of isotope, summed:

$$h = \sum_i c_i h_i, \quad c_i(\text{past}) = c_i 2^{t/t_{1/2}}$$

- Today: $\approx 4.9 \times 10^{-12} \text{ W kg}^{-1}$ of BSE rock $\Rightarrow \approx 20 \text{ TW}$.
- The same $2^{t/t_{1/2}}$ law runs every clock: at 10 Myr, ^{26}Al is 13.9 half-lives down, a factor 1.5×10^4 .

Two clocks, two eras

^{26}Al ($t_{1/2} = 0.72 \text{ Myr}$) out-heats everything for the first few Myr, then dies: planetesimals melt on the short clock, planets stay warm on the long one.

The isotope table is the whole calculation: concentrations, half-lives, heat rates. Everything else is bookkeeping.

Conduction fails, convection takes over

Conductive cooling time and convective vigour:

$$\tau = \frac{L^2}{\kappa},$$

$$Ra = \frac{\rho g \alpha \Delta T D^3}{\kappa \mu}$$

- Moon: $\tau \approx 10^{11}$ yr $\gg$ age.
- Only bodies ≤ 400 km cool conductively in 4.57 Gyr.
- Mantle: $Ra \approx 5 \times 10^7 \gg Ra_c \approx 1708$.

How much faster is convection?

The Nusselt number $Nu = (Ra/Ra_c)^{1/3}$ is the factor by which convection beats conduction. Since $Nu \propto \Delta T^{1/3}$, the flux scales as $\Delta T^{4/3}$: hotter mantles shed heat disproportionately fast.

L^2 decides who conducts; Ra decides who convects; Nu says how vigorously.

A minimal thermal evolution model

One line of energy conservation:

$$C \frac{dT}{dt} = H(t) - Q(T)$$

- $H(t)$: radiogenic heating, a decaying exponential.
- $Q(T) \propto T^{4/3}$: convective loss (from $Nu \propto Ra^{1/3}$).
- You read peak, endpoint, and Urey ratio off the printed solution curve.

What the curve does

Start hot but under-radiating: the interior *warms* until Q catches H (~1 Gyr), then tracks the dying heat source down. Today: Urey ratio $H/Q \approx 0.4$.

Planets self-regulate: the $T^{4/3}$ thermostat ties the heat loss to the heat supply.

Building a core and dating it

One formula, one story:

$$\frac{C^{\text{sil}}}{C^{\text{bulk}}} = \frac{1}{1 + x(D - 1)}$$

- Siderophile elements ($D \gg 1$) vanish into the core.
- Their leftover excess in the mantle records the **late veneer**.

Core formation is chemically selective and datable: the mantle keeps both the receipt and the clock.

The Hf-W stopwatch

$^{182}\text{Hf} \rightarrow ^{182}\text{W}$, $t_{1/2} = 8.9$ Myr: Hf stays in the mantle, W leaves with the metal. The mantle's $\epsilon^{182}\text{W}$ excess dates core formation, until the parent goes extinct at ~45 Myr.

Dynamos and shields

The shield and its lifetime:

$$r_{\text{mp}} = R_p \left(\frac{B_0^2}{\mu_0 \rho_{\text{sw}} v_{\text{sw}}^2} \right)^{1/6}, \quad \tau_{\text{ohm}} = \frac{L^2}{\eta}$$

- For Ganymede's 300 km shell, $\tau_{\text{ohm}} \sim$ kyr.
- So an observed field is regenerated *now*: no fossil survives.

The sixth root

A tenfold solar-wind surge moves Earth's magnetopause only from 10 to $6.8 R_{\oplus}$. Weak dependence is the shield's strength.

The pressure balance says how far the shield reaches; ohmic decay says a field must be driven today.

Working the worksheet

- **Show your algebra.** The exam asks for the steps, not only the number.
- Carry **4 significant figures** through intermediate steps; round at the end.
- Use the **checkpoints** ("show that ...") to verify your work and to keep moving if a step resists.
- Qualitative parts want **2 to 3 sentences** of physics, not an essay.
- Problem 3 prints the model's solution curves: read the peak, the endpoint, and the Urey ratio straight off them.

Worksheet and full solutions: ips.formingworlds.space/worksheets.html.
We discuss questions, not presentations of the solutions.